Certified Penetration Testing Professional

Hashcat Cheat Sheet

Hashcat

Source: *https://hashcat.net*

Hashcat is the advanced password recovery utility and it can generate combinations of brute-force attack possibilities.

Syntax
`hashcat [options]... hash|hashfile|hccapxfile [dictionary|mask|directory]...`

1. Hashcat Options

Options	Description
-m or --hash-type	Hash-type
-a or --attack-mode	Attack-mode
-V	Print version

or **--version**	
-h **or** **--help**	Print help
--quiet	Suppress output
--hex-charset	Assume charset is given in hex
--hex-salt	Assume salt is given in hex
--hex-wordlist	Assume words in wordlist are given in hex
--force	Ignore warnings
--deprecated-check-disable	Enable deprecated plugins
--status	Enable automatic update of the status screen
--status-json	Enable JSON format for status output
--status-timer	Sets seconds between status screen updates to X
--stdin-timeout-abort	Abort if there is no input from stdin for X seconds
--machine-readable	Display the status view in a machine-readable format
--keep-guessing	Keep guessing the hash after it has been cracked
--self-test-disable	Disable self-test functionality on startup
--loopback	Add new plains to induct directory
--markov-hcstat2	Specify hcstat2 file to use
--markov-disable	Disables markov-chains, emulates classic brute-force
--markov-classic	Enables classic markov-chains
--markov-inverse	Enables inverse markov-chains
-t **or** **--markov-threshold**	Threshold X when to stop accepting new markov-chains
--runtime	Abort session after X seconds of runtime
--session	Define specific session name
--restore	Restore session from --session
--restore-disable	Disable restored file
--restore-file-path	Specific path to restore file
-o	Define outfile for recovered hash

or **--outfile**	
--outfile-format	Define outfile-format X for recovered hash
--outfile-autohex-disable	Disable the use of $HEX[] in output plains
--outfile-check-timer	Sets seconds between outfile checks to X
--wordlist-autohex-disable	Disable the conversion of $HEX[] from the wordlist
-p **or** **--separator**	Separator char for hashlists and outfile
--stdout	Do not crack a hash, instead print candidates only
--show	Compare hashlist with potfile; show cracked hashes
--left	Compare hashlist with potfile; show uncracked hashes
--username	Enable ignoring of usernames in hashfile
--remove	Enable removal of hashes once they are cracked
--remove-timer	Update input hash file each X seconds
--potfile-disable	Do not write potfile
--potfile-path	Specific path to potfile
--encoding-from	Force internal wordlist encoding from X
--encoding-to	Force internal wordlist encoding to X
--debug-mode	Defines the debug mode
--debug-file	Output file for debugging rules
--induction-dir	Specify the induction directory to use for loopback
--logfile-disable	Disable the logfile
--hccapx-message-pair	Load only message pairs from hccapx matching X
--nonce-error-corrections	The BF size range to replace AP's nonce last byte
--keyboard-layout-mapping	Keyboard layout mapping table for special hash-modes
--truecrypt-keyfiles	Keyfiles to use, separated with commas
--veracrypt-keyfiles	Keyfiles to use, separated with commas
--veracrypt-pim-start	VeraCrypt personal iterations multiplier start
--veracrypt-pim-stop	VeraCrypt personal iterations multiplier stop

-b **or** **--benchmark**	Run benchmark of selected hash-modes
--benchmark-all	Run benchmark of all hash-modes
--speed-only	Return expected speed of the attack, then quit
--progress-only	Return ideal progress step size and time to process
-c **or** **--segment-size**	Sets size in MB to cache from the word file to X
--bitmap-min	Sets minimum bits allowed for bitmaps to X
--bitmap-max	Sets maximum bits allowed for bitmaps to X
--cpu-affinity	Locks to CPU devices, separated with commas
--example-hashes	Show an example hash for each hash-mode
--hook-threads	Sets number of threads for a hook
-I **or** **--backend-info**	Show system/environment/backend API info
-d **or** **--backend-devices**	Backend devices to use, separated with commas
-D **or** **--opencl-device-types**	OpenCL device-types to use, separated with commas
-O **or** **--optimized-kernel-enable**	Enable optimized kernels (limits password length)
-w, --workload-profile	Enable a specific workload profile, see pool below
-n **or** **--kernel-accel**	Manual workload tuning, set outerloop step size to X
-u **or** **--kernel-loops**	Manual workload tuning, set innerloop step size to X
-T **or**	Manual workload tuning, set thread count to X

`--kernel-threads`	
`--spin-damp`	Use CPU for device synchronization, in percent
`--hwmon-disable`	Disable temperature and fanspeed reads and triggers
`--hwmon-temp-abort`	Abort if temperature reaches X degrees Celsius
`--scrypt-tmto`	Manually override TMTO value for scrypt to X

`-s`
`or`
`--skip`
	Skip X words from the start

`-l`
`or`
`--limit`
	Limit X words from the start + skipped words

`--keyspace`	Show keyspace base:mod values and quit

`-j`
`pr`
`--rule-left`
	Single rule applied to each word from left wordlist

`-k`
`or`
`--rule-right`
	Single rule applied to each word from right wordlist

`-r`
`or`
`--rules-file`
	Multiple rules applied to each word from wordlists

`-g`
`or`
`--generate-rules`
	Generate X random rules

`--generate-rules-func-min`	Force min X functions per rule
`--generate-rules-func-max`	Force max X functions per rule
`--generate-rules-seed`	Force RNG seed set to X

`-1`
`or`
`--custom-charset1`
	User-defined charset 1

`-2`
`or`
`--custom-charset2`
	User-defined charset 2

`-3` `or`	User-defined charset 3

--custom-charset3	
-4 **or** **--custom-charset4**	User-defined charset 4
-i **or** **-- --increment**	Enable mask increment mode
--increment-min	Start mask incrementing at X
--increment-max	Stop mask incrementing at X
-S **or** **--slow-candidates**	Enable slower (but advanced) candidate generators
--brain-server	Enable brain server
-z **or** **--brain-client**	Enable brain client, activates -S
--brain-client-features	Define brain client features, see below
--brain-host	Brain server host (IP or domain)
--brain-port	Brain server port
--brain-password	Brain server authentication password
--brain-session	Overrides automatically calculated brain session
--brain-session-whitelist	Allow given sessions only, separated with commas

2. Hashcat-utils

Utilities	Description
cap2hccapx **Syntax: ./cap2hccapx.bin input.pcap output.hccapx [filter by essid] [additional network essid:bssid]**	Specifies a network name (ESSID) to filter out unwanted networks
cleanup-rules **Syntax: ./cleanup-rules.bin mode**	Strips rules from STDIN that are not compatible with required platform
combinator **Syntax: ./combinator.bin file1 file2**	Each word from file2 is appended to each word from file1 which is then printed to STDOUT

`combinator3`	Accepts three files as input, that produces the output combining all three lists
`combipow`	Produces all unique combinations from a shortlist of inputs
`cutb` `Syntax: ./cutb.bin offset [length] < infile > outfile`	It will cut the specific prefix or suffix length off the existing words in a list and pass it to STDOUT
`expander`	Each word going into STDIN is parsed and split into all its single chars then sent to STDOUT
`gate` `Syntax: ./gate.bin mod offset < infile > outfile`	Each wordlist going into STDIN is parsed split into equal sections and passed to STDOUT
`generate-rules` `Syntax: ./generate-rules.bin number [seed]`	Stand-alone utility to generate random rules
`hcstatgen` `Syntax: ./hcstatgen.bin out.hcstat < infile`	Generates .hcstat files for use with older hashcats
`hcstat2gen` `Syntax: ./hcstat2gen.bin outfile < dictionary`	Generates custom Markov statistics for use with hashcat's `--markov-hcstat` parameter
`keyspace` `Syntax: ./keyspace.bin [options] mask`	Calculates keyspace in a hashcat-aware manner
`len` `Syntax: ./len.bin min max < infile > outfile`	Each word going into STDIN is parsed for its length and passed to STDOUT if it matches a specified word-length range
`mli2` `Syntax: ./mli2.bin infile mergefile`	Merges two lists
`morph` `Syntax: ./morph.bin dictionary depth width pos_min pos_max`	Generates insertion rules for the most frequent chains of characters
`./permute_exist.bin word < infile > outfile`	Each word going into STDIN is parsed and run through “The Countdown QuickPerm Algorithm”
`prepare`	Made as dictionary optimizer for Permutation attack

req-include	Each word going into STDIN is parsed and passed to STDOUT if it matches a specified password group criterion
req-exclude	Excludes words that match specific criteria
rli **Syntax: rli infile outfile removefiles...**	Compares a single file against another file removing all duplicates
splitlen **Syntax: ./splitlen.bin outdir < infile**	It is designed to be a dictionary optimizer for the now deprecated oclHashcat
strip-bsn	Strips all \0 bytes from stdin
strip-bsr	Strips all \r bytes from stdin
tmesis	Take wordlist and produce insertion rules
tmesis-dynamic **Syntax: ./tmesis-dynamic.pl substring wordlist1.txt wordlist2.txt**	Take 2 wordlists and produces new one, using a user-defined substring as a “key”

3. Maskprocessor

Syntax
./mp64.bin [options]... mask

Options	
-V **or** **--version**	Print version
-h **or** **--help**	Print help
-I **or** **--increment=NUM:NUM**	Enable increment mode. 1st NUM=start, 2nd NUM=stop
--combinations	Calculate number of combinations
--hex-charset	Assume charset is given in hex
-q	Maximum number of multiple sequential characters

or **--seq-max=NUM**	
-r **or** **--occurrence-max=NUM**	Maximum number of occurrences of a character
-s **or** **--start-at=WORD**	Start at specific position
-l **or** **--stop-at=WORD**	Stop at specific position
-o **or** **--output-file=FILE**	Output-file

4. Statsprocessor

Syntax
./sp64.bin [options]... hcstat-file [filter-mask]

Options	
-V **or** **--version**	Print version
-h **or** **--help**	Print help
--pw-min=NUM	Start incrementing at NUM
--markov-disable	Emulates maskprocessor output
--markov-classic	No per-position tables
--threshold=NUM	Filter out chars after NUM chars added set to 0 to disable
--combinations	Calculate number of combinations
--hex-charset	Assume charset is given in hex
-s **or**	Skip number of words (for restore)

--skip=NUM	
-l **or** **--limit=NUM**	limit number of words (for distributed)
-o **or** **--output-file=FILE**	Output-file

5. Kwprocessor

Syntax
./kwp [options]... basechars-file keymap-file routes-file

Options	
-V **or** **--version**	Print version
-h **or** **--help**	Print help
-o **or** **--output-file**	Output file
-b **or** **--keyboard-basic**	Include characters reachable without holding shift or altgr
-s **or** **--keyboard-shift**	Include characters reachable by holding shift
-a **or** **--keyboard-altgr**	Include characters reachable by holding altgr
-z **or** **--keyboard-all**	Shortcut to enable all --keyboard-* modifier
-1 **or** **--keywalk-south-west**	Include routes heading diagonal south-west

-2 **or** **--keywalk-south**	Include routes heading straight south
-3 **or** **--keywalk-south-east**	Include routes heading diagonal south-east
-4 **or** **--keywalk-west**	Include routes heading straight west
-5 **or** **--keywalk-repeat**	Include routes repeating character
-6 **or** **--keywalk-east**	Include routes heading straight east
-7 **or** **--keywalk-north-west**	Include routes heading diagonal north-west
-8 **or** **--keywalk-north**	Include routes heading straight north
-9 **or** **--keywalk-north-east**	Include routes heading diagonal north-east
-0 **or** **--keywalk-all**	Shortcut to enable all --keywalk-* directions
-n **or** **--keywalk-distance-min**	Minimum allowed distance between keys
-x **or** **--keywalk-distance-max**	Maximum allowed distance between keys